Practical 13: Spinner and ListView Adapters

Source: MODULE 2.pdf | Source pages: 30–34

Aim

To develop and test an Android application demonstrating spinner and listview adapters.

Steps to Perform the Practical

1. Create a project named AdapterDemo.

2. Add a Spinner and ListView to activity_main.xml.

3. In MainActivity.java, define the cities array shown in the practical.

4. Create ArrayAdapter objects for the Spinner and ListView and attach them to the views.

5. Add selection and item-click listeners that display Toast messages.

6. Run the app, choose different cities from the Spinner, and tap ListView items.

Result

The application is run and the expected behavior is verified according to the practical instructions and code.

Note: Use the package name and resource names from the source code consistently. If Android Studio reports a missing resource or class, check spelling and ensure all files are in the correct folders.

6. Develop an Android application using Adapters to populate Spinner or ListView dynamically.

Android Spinner is a view similar to the dropdown list which is used to select one option from the list of options. It provides an easy way to select one item from the list of items and it shows a dropdown list of all values when we click on it. The default value of the android spinner will be the currently selected value and by using **Adapter** we can easily bind the items to the spinner objects. Generally, we populate our Spinner control with a list of items by using an **ArrayAdapter** in our Kotlin/Java file.

Project Name: AdapterDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:orientation="vertical"

   android:padding="20dp">

   <Spinner

       android:id="@+id/spinner"

       android:layout_width="match_parent"

       android:layout_height="wrap_content"/>

   <ListView

       android:id="@+id/listView"

       android:layout_width="match_parent"

       android:layout_height="wrap_content"

       android:layout_marginTop="20dp"/>

</LinearLayout>
```

File Name: MainActivity.java

```
package com.example.adapterdemo;


import android.os.Bundle;

import android.view.View;

import android.widget.AdapterView;

import android.widget.ArrayAdapter;

import android.widget.ListView;

import android.widget.Spinner;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


   Spinner spinner;

   ListView listView;


   String[] cities = {

           "Mumbai",

           "Pune",

           "Delhi",

           "Chennai",

           "Hyderabad"

   };
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);


        spinner = findViewById(R.id.spinner);
        listView = findViewById(R.id.listView);

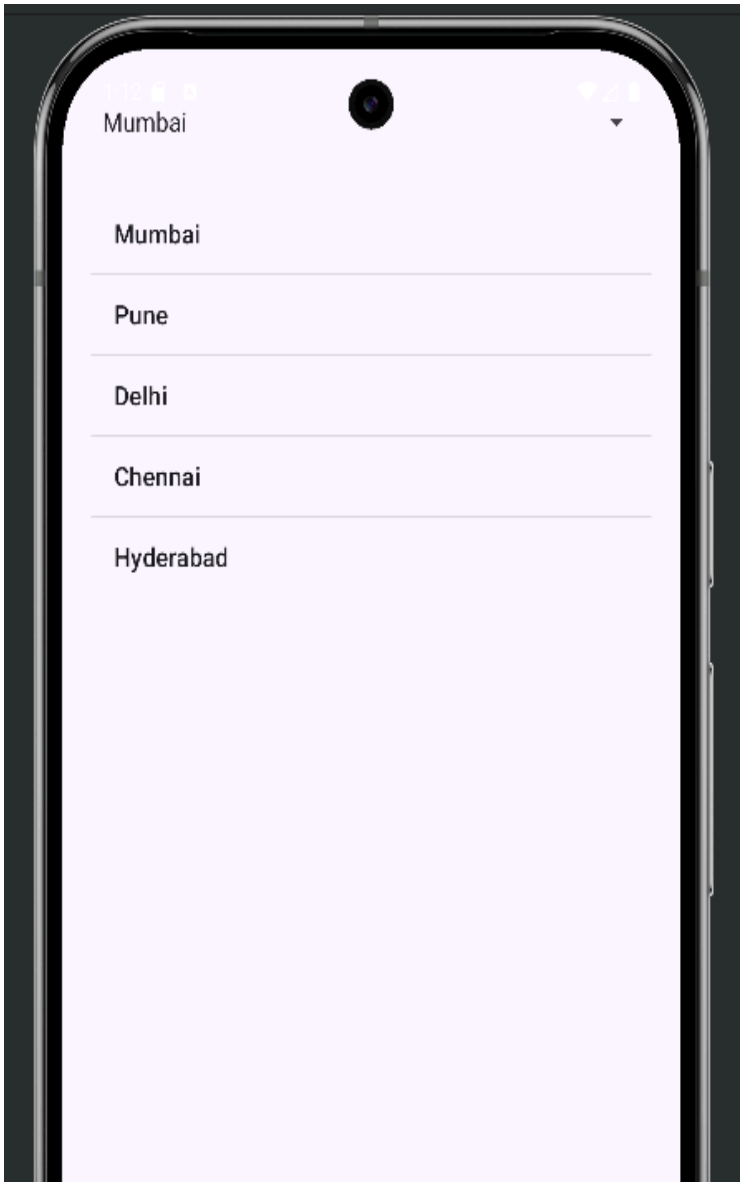

        // Spinner Adapter
        ArrayAdapter<String> spinnerAdapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_spinner_dropdown_item,
                cities
        );


        spinner.setAdapter(spinnerAdapter);
        // ListView Adapter
        ArrayAdapter<String> listAdapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                cities
        );
        listView.setAdapter(listAdapter);
        // Spinner Item Selection
        spinner.setOnItemSelectedListener(new
AdapterView.OnItemSelectedListener() {
```

```
            @Override
            public void onItemSelected(AdapterView<?> parent, View view,
                                       int position, long id) {


                Toast.makeText(
                        MainActivity.this,
                        "Selected: " + cities[position],
                        Toast.LENGTH_SHORT
                ).show();
            }
            @Override
            public void onNothingSelected(AdapterView<?> parent) {


            }
        });
        // ListView Item Click
        listView.setOnItemClickListener((parent, view, position, id) -> {
            Toast.makeText(
                    MainActivity.this,
                    "Clicked: " + cities[position],
                    Toast.LENGTH_SHORT
            ).show();
        });
    }
}
```


Mumbai
Mumbai
Pune
Delhi
Chennai
Hyderabad